TAXWISE

Ask. Learn. File.

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Problem

Most students file their first tax return with no formal instruction, and the official information they need is spread across dozens of long, dense documents.

- **Fragmented sources.** The rules for one situation (a first W-2 job plus a scholarship) can span Form 1040 instructions, several IRS publications and a state revenue site.
- **Jargon barrier.** Terms like adjusted gross income, standard deduction and withholding are assumed knowledge, not explained.
- **Generic answers.** Search results and blogs rarely match a student's exact situation, and are often outdated or written to sell tax software.
- **Untrustworthy AI.** General chatbots answer confidently but can invent rules or cite the wrong tax year, with no way to check the source.

Target user: a college student filing a federal return for the first or second time.

Motivation: a tool that answers tax questions in plain language, grounded in and linked to the official source text, would make taxes learnable instead of intimidating. It is also a strong test of vibe coding, because a wrong answer is costly and easy to verify against the source.

Solution

TaxWise is a desktop app that teaches personal taxes through a retrieval-augmented generation (RAG) chatbot grounded only in official IRS and state documents, with every answer linked to its source passage.

Core features:

1. **Cited Q&A chat.** The user asks a question in plain English; the system retrieves the relevant passages and answers with inline citations that open the exact source page.
    - **Plain-language mode.** A toggle that rewrites answers at a beginner reading level and defines each tax term on hover.
2. **"Explain my form" walkthrough.** The user uploads or picks a sample W-2 or 1098-T, and the app explains each box and where it goes on Form 1040.
3. **Scenario calculator.** A simple estimator (income, filing status, dependency) that computes tax with deterministic code, not the LLM, so numbers are always correct.
4. **Learning quizzes.** Short generated quizzes on topics the user has asked about, to reinforce concepts.

Guardrails: the assistant says "I don't know" when retrieval finds nothing relevant, labels the tax year of every source, and shows a disclaimer that it is an educational tool, not professional tax advice.

System Design/Architecture

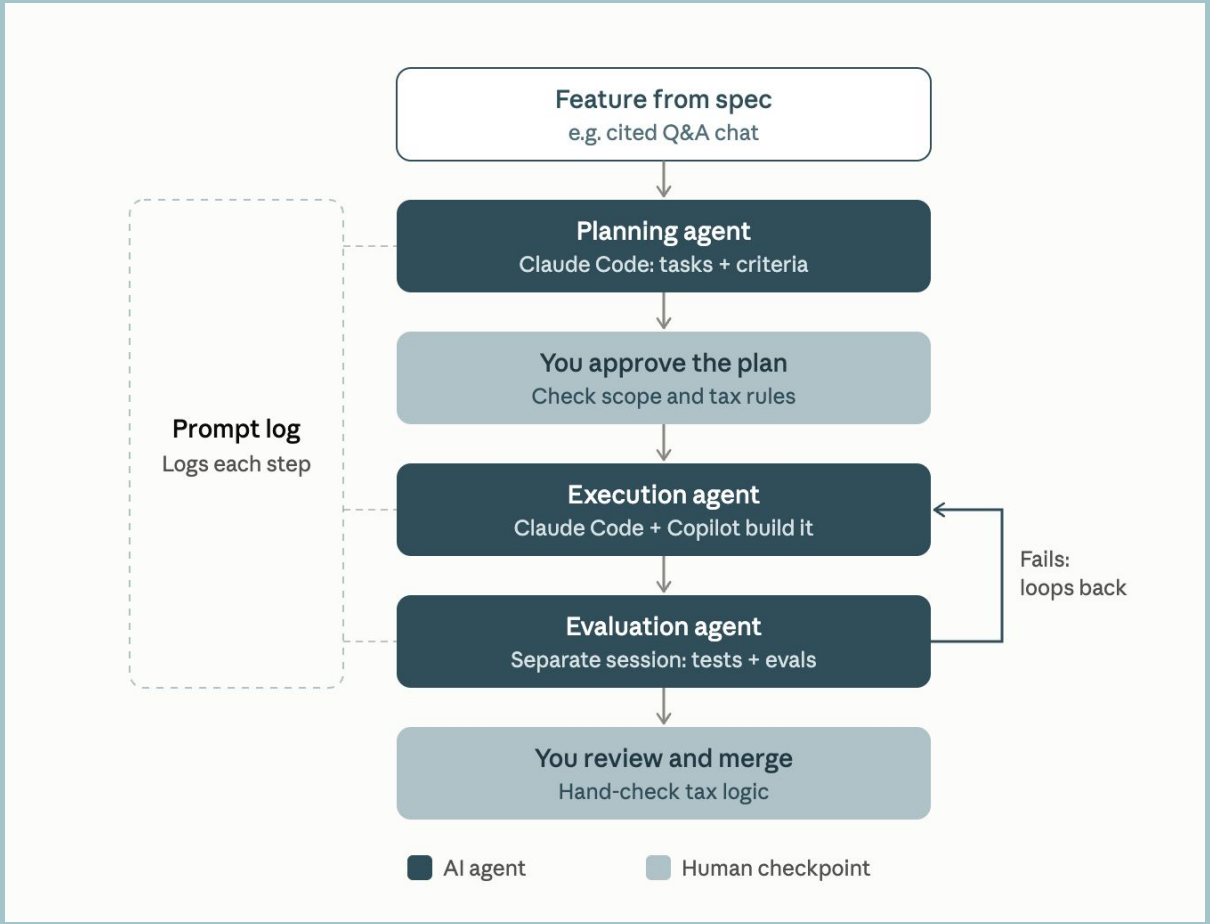
Two pipelines: an offline ingestion pipeline builds the knowledge base once, and an online query pipeline runs on every question.

- **Offline ingestion:** IRS PDFs and state guides → parse text (PdfPig) → chunk by section + metadata → embed (Voyage AI) → vector DB (Qdrant / pgvector)
- **Online query:** MAUI Blazor UI → ASP.NET Core API → Semantic Kernel (vector search top 8, rerank to top 4) → Claude answers only from the passages, citing each claim → clickable citations in the UI
- **Tax calculator:** a C# class computes every tax amount; the LLM never does math, it only explains the calculator's output.

Knowledge base (v1): Publications 17, 970, 501 and 519, plus the Form 1040, W-2 and 1098-T instructions. Each chunk stores its source, section heading, page number and tax year.

Stack: Visual Studio · ASP.NET Core Web API · Semantic Kernel · PdfPig · Qdrant or pgvector · Voyage AI embeddings · Claude via Anthropic's C# SDK · xUnit tests · Azure App Service

AI Development Strategy



AI Integration: Cited Q&A Chat

User asks a tax question in plain English and gets an answer with clickable citations that open the exact page of the official IRS or state document it came from to give user confidence in the material's credibility.

AI's Role

Claude writes the answer to the user's question in plain English

It writes only from the 4 retrieved passages, never from its own memory

How it works

The question is embedded, matched to the top 8 chunks, then reranked to the best 4

Claude cites a passage ID for each claim, and each ID becomes a clickable link to the source page

Plain Language Mode

A toggle that rewrites the answer at a beginner reading level and defines tax terms

It changes only the writing instructions, so the passages and facts stay the same

Guardrail

If the best match scores under a set similarity threshold, we skip Claude and return "I don't know"

The API checks that every cited ID exists in the 4 passages sent, and every source shows its tax year

TaxWise

- Ask a question
- Explain my form
- Scenario calculator
- Learning quizzes

KNOWLEDGE BASE

Pubs 17, 970, 501, 519
Form 1040, W-2, 1098-T instructions
Tax year 2025

Scholarship and housing money

Plain-language mode

My scholarship paid my tuition and also gave me $1,200 for housing. Do I have to pay tax on it?

Searched 8 passages, kept the 4 most relevant

Part of it, yes. The money that went straight to your **tuition and required course fees** is usually not taxed, as long as you are working toward a degree.

1 Pub 970 · Ch. 1 · p. 6 · TY 2025

The **$1,200 for housing** is different. Money used for room and board counts as income, so it gets added to your taxable income on your return.

2 Pub 970 · Ch. 1 · p. 7 · TY 2025

Where do I report this?

Quiz me on scholarships

Ask a tax question

Ask

SOURCE PASSAGE 2

Publication 970

Tax Benefits for Education

Tax year 2025 | Chapter 1 | Page 7

Scholarships and fellowships

> [Excerpt of the retrieved passage appears here. The sentence the answer relied on is highlighted.]
>
> Scholarship amounts used for room, board, or other costs that are not required for enrollment are generally treated as income.
>
> [Rest of the passage continues.]

Open page 7 in the PDF

AI Integration: Explain My Form

Walks you through a W-2 or 1098-T box by box, explaining what each number means and where it goes on your Form 1040. *User uploads a tax form and the inputs are explained

AI's Role

Claude explains what one box on a W-2 or 1098-T means in plain language

It also says where that number goes on Form 1040

How it works

The user uploads a PDF or or picks a sample.

PdfPig reads text PDFs directly, and Claude returns box numbers and values as structured data.

Scope

The prototype shows sample forms. Reading an uploaded form is a stretch goal, since pulling box values out of a scan is its own problem.

Guardrail

Each explanation is tied to one box, one form and one tax year, so the AI never guesses what a box means

If no passage matches the box, the app shows "I don't know" instead of an explanation

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Pubs 17, 970, 501, 519
Form 1040, W-2, 1098-T instructions
Tax year 2025

Explain my form

Sample W-2 | Sample 1098-T | Upload mine

Form W-2

Wage and Tax Statement · sample

c · Employer name
[Sample Employer Inc.]

1 · Wages, tips, other comp.	2 · Federal income tax withheld
$[0,000.00]	$[000.00]
3 · Social security wages	4 · Social security tax withheld
$[0,000.00]	$[000.00]
5 · Medicare wages	6 · Medicare tax withheld
$[0,000.00]	$[000.00]
15 · State	17 · State income tax
CO	$[000.00]

Tap any box to see what it means.

1 Wages, tips, other compensation

This is the total your employer paid you for the year that counts as taxable income. It is the main number you start with when you fill out your return.

WHERE IT GOES
Form 1040 · line 1a

1 W-2 instructions · Box 1 · TY 2025
2 Form 1040 instructions · Line 1a · TY 2025

NEXT UP

Box 2 · Federal tax withheld

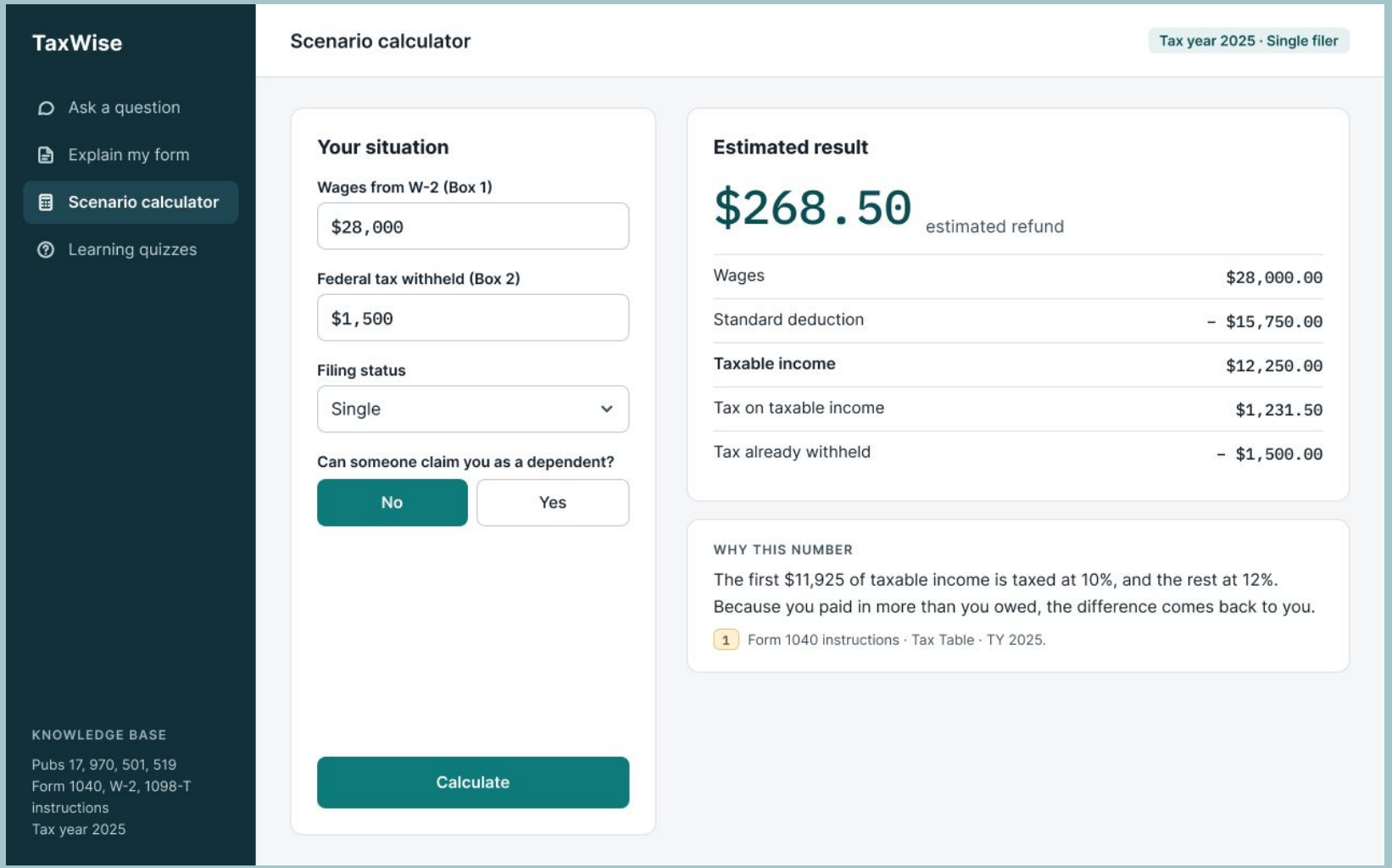
The tax already paid for you. It can come back as a refund.

Walk me through the next box

AI Integration: Scenario Calculator

Estimates your tax or refund from your income, filing status and dependency, with every number computed by code and explained step by step.

AI's Role	How it works	Why this number	Guardrail
Claude explains the result in words and does no math It writes the "why this number" paragraph	C# class computes taxable income, tax and refund from income, filing status and dependency The finished numbers are passed to Claude as fixed data, along with the retrieved tax table passage	The result shows each step (wages, standard deduction, taxable income, tax, withholding) in a table next to the explanation	A check compares every dollar amount in Claude's explanation to the calculator's output and blocks any mismatch


TaxWise
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Scenario calculator
Learning quizzes
KNOWLEDGE BASE
Pubs 17, 970, 501, 519
Form 1040, W-2, 1098-T instructions
Tax year 2025
Scenario calculator
Tax year 2025 · Single filer
Your situation
Wages from W-2 (Box 1)
$28,000
Federal tax withheld (Box 2)
$1,500
Filing status
Single
Can someone claim you as a dependent?
No
Yes
Calculate
Estimated result
$268.50
estimated refund
Wages $28,000.00
Standard deduction – $15,750.00
Taxable income $12,250.00
Tax on taxable income $1,231.50
Tax already withheld – $1,500.00
WHY THIS NUMBER
The first $11,925 of taxable income is taxed at 10%, and the rest at 12%. Because you paid in more than you owed, the difference comes back to you.
1 Form 1040 instructions · Tax Table · TY 2025.

AI Integration: Learning Quizzes

Turns the topics you've asked about into short quizzes with explanations and citations, so the concepts actually stick.

AI's Role

Claude writes multiple-choice questions on topics the user has asked about

Each question comes with an answer key and a short explanation

How it works

Each question is generated from one retrieved passage, returned as structured JSON (question, 4 options, correct answer, explanation, passage ID)

The explanation cites the passage, so a user who gets it wrong can open the source

Answer Sources

After each answer, the user sees whether they were right, why, and a link to the source page

Progress shows as "Question 2 of 5," drawn from the topics the user asked about

Guardrail

Questions come only from retrieved passages, never from the model's general knowledge

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Tax year 2025

Quiz · Scholarships

Question 2 of 5

GENERATED FROM YOUR QUESTIONS ABOUT SCHOLARSHIPS

A scholarship gives you $1,200 that you spend on rent. How is that money treated on your return?

A. It is never taxed

B. It counts as taxable income

C. It reduces your standard deduction

D. It is taxed only if you are over 24

Correct

Scholarship money used for housing is not covered by the tuition exclusion, so it is added to your income.

1 Pub 970 · Ch. 1 · p. 7 · TY 2025

Next question